

Rat-Bite Fever

Lesions are most prominent on the palms, soles, and around joints but may become generalized, resembling measles. Notably, when the rash appears, the rat bite has usually healed and only rarely shows evidence of infection, inflammation, or regional adenopathy. A week or so after resolution of the disease, the palms and soles exhibit characteristic desquamation that may resemble Kawasaki disease. In contrast, the bite producing spirillary fever is usually tender, red, indurated, or ulcerated when the rash appears. Cutaneous lesions are larger and are prominent on the abdomen, thus resembling the rose spots of typhoid.

Related Physical Findings

Within a week of the febrile symptoms, arthritis usually develops, predominantly involving large joints. The arthritis may present as an asymmetric polyarthritis resembling rheumatoid arthritis but is rarely suppurative. Regional lymphadenopathy may be present.

Laboratory Findings

Differential Diagnosis

Treatment

The treatment of choice for someone who becomes ill after a rat bite is amoxicillin/clavulanic acid. In patients allergic to penicillins, doxycycline is acceptable. Once *Streptobacillus moniliformis* is identified, penicillin is the drug of choice. In general, rat bites should be cleansed promptly and thoroughly. Tetanus immunization status should be checked. Although there are no established guidelines, one might consider a prompt prophylactic course of penicillin or doxycycline after any rat bite.

Prognosis, Clinical Course, and Complications

Prevention

Seal Finger

Etiology and Epidemiology

Seal finger is an occupational zoonosis that occurs only after direct contact with seals, sea lions, and similar marine mammals (order Pinnipedia), transmitted via bites or through open skin after handling seals or their carcasses. It is characterized by intensely painful red nodules, usually on the distal phalanx. Untreated, it can progress to tenosynovitis, osteitis, and joint destruction. Although not fully confirmed, seal finger appears to be caused by a group of *Mycoplasma* species that are part of the normal oropharyngeal flora of pinnipeds.

At a Glance

- Occupational zoonosis after contact with seals or sea lions
- Occurs worldwide in marine environments
- Usually presents as an exquisitely painful nodule on the distal phalanx of the finger that may progress to tenosynovitis or joint destruction
- The pathogen is likely a marine *Mycoplasma* found in the seal's normal oropharyngeal flora
- The infection responds quickly to tetracyclines

Clinical Findings

People at risk include hunters, zookeepers, and marine biologists. Seal finger is common in coldwater coastal areas around Scandinavia, Greenland, Newfoundland, and similar environments in the Southern Hemisphere. Most cases occur during the spring sealing season. The incubation period is usually

approximately 4 days, though it can rarely extend up to 2 weeks. A furuncle-like lesion appears at the inoculation site, followed by severe pain, marked swelling, and stiffness.

Differential Diagnosis

Treatment

Seal finger does not respond to β -lactam antibiotics, which are frequently prescribed empirically for cellulitis caused by *Staphylococcus*, *Streptococcus*, or *Erysipelothrix*. The lack of response to β -lactams further supports the diagnosis of seal finger.